

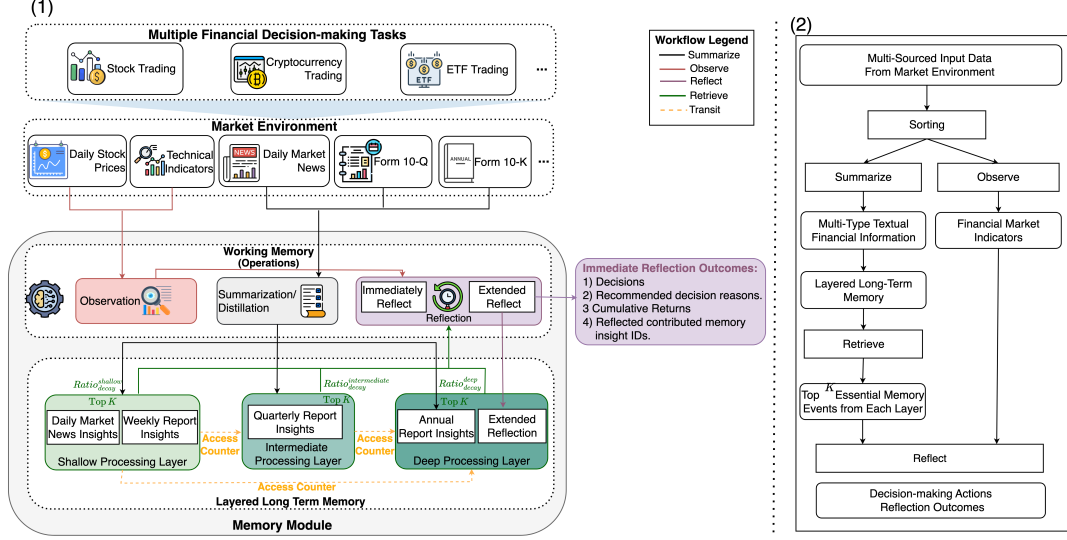


Figure 2: (1) The language agent’s memory module is crafted to interact with the market environment to conduct various financial decision-making tasks. It contains two core components – Working Memory and Layered Long-term Memory. (2) The outline of the agent’s decision-making workflow for retrieving critical memory events and market observations to inform specific investment decisions.

This POMDP contains: (1) a state space $\mathcal{X} \times \mathcal{Y}$ where $\mathcal{X}$ is the observable component and $\mathcal{Y}$ is unobservable component of the financial market; (2) the action space of the agent is $\mathcal{A}$, which is modeled as $\{“Buy”, “Sell”, “Hold”\}$; (3) the reward function $\mathcal{R}(o, b, a) : \mathcal{X} \times \mathcal{Y} \times \mathcal{A} \rightarrow \mathbb{R}$ uses daily profit & loss (PnL) as the output; (4) the observation process $\{O_t\}_{t \in \mathbb{T}} \subseteq \mathcal{X}$ is a multi-dimensional process (5) the reflection process $\{B_t\}_{t \in \mathbb{T}} \subseteq \mathcal{Y}$ represents the agent’s self-reflection, which is updated from B_t to B_{t+1} on daily basis (Griffiths et al., 2023); (6) the action $A_t \sim \pi(\cdot | \text{prompt})$ represents the way to make investment decision driven by the language conditioned policy π . By denoting daily profit & loss (PnLs) by $R_t^\pi = \mathcal{R}(O_t, B_t, A_t)$ and the set of all admissible language conditioned policies as $\Pi = \{\pi(\cdot | \text{prompt})\}$, the optimization objective of financial decision-making task is then:

$$\max_{\pi \in \Pi} \mathbb{E} \left[\sum_{t \in \mathbb{T}} \alpha^t R_t^\pi \right] \quad (1)$$

3 InvestorBench

Here we introduce the detailed architecture of InvestorBench, as illustrated in Figure 1.

3.1 Benchmark Composition

INVESTORBENCH is organized into four main components: (1) **Data Sources and Market Environments**: INVESTORBENCH utilizes a wide range of open-source data and incorporates third-party APIs,

such as Yahoo Finance and SEC EDGAR, to create a comprehensive, multi-modal market environment data warehouse. (2) **LLM Agent**: INVESTORBENCH includes an advanced LLM-based agent equipped with modules for Brain, Perception, Profile, Memory, and Action. This agent is enhanced with external tools (such as tabular data readers and API callers) and data operations (including vector database management, information reinforcement, and retrieval). (3) **Financial Decision-Making Tasks**: INVESTORBENCH offers three distinct financial decision-making tasks, differentiated by their asset types. (4) **Evaluation Metrics**: The efficacy of all tasks within INVESTORBENCH is evaluated using a set of standard metrics in the quantitative finance field, providing a thorough evaluation of the decision-making capabilities of the LLM-based agent.

3.2 Trading Environments

We release three datasets, each curated from diverse sources, to construct tailored financial market environments for specific tasks. Our objective is to address the current gap in evaluation environments for financial decision-making agent frameworks and to offer a fully open platform for the comprehensive assessment of agents across various tasks. Below, we introduce each environment, categorized by task type, detailing its scope and the data sources it incorporates.

Stock market environment integrates information from multiple sources, including: 1) Daily stock open, high, low, close, and volume (OHLCV) data acquired from Yahoo Finance. 2) Summarized insights from company quarterly and annual reports (Form 10-Q and 10-K) downloaded from the SEC EDGAR database. 3) News articles for seven stocks collected daily between 2020-07-01, and 2021-05-06. The news data for four of these companies—Microsoft Corporation (MSFT), Johnson & Johnson (JNJ), UVV Corporation (UVV), and Honeywell International Inc. (HON)—are randomly selected from the pool with the most new records (over five hundred) from the open-access dataset provided by Zhou et al. (Zhou et al., 2021), while the news data for the remaining three companies—Tesla, Inc. (TSLA), Apple Inc. (AAPL), and NIO Inc. (NIO)—are obtained from Refinitiv Real-Time News, which primarily contains high-quality news information from Reuters. 4) The sentiment categories (*‘positive’*, *‘negative’*, *‘neutral’*) assigned to each news record are generated by gpt-3.5-turbo-0125.

Cryptocurrency market environment encompasses 1) the daily stock open-high-low-close-volume (OHLCV) acquired from CoinMarketCap; 2) the multisource cryptocurrency news data collected from cryptonews, cryptopotato, and cointelegraph (Vanhoucke, 2023); 3) news spanning from 2023-02-13 to 2023-11-05 collected by (Zhou et al., 2021) in daily frequency. 4) The sentiment categories generated by the same means.

ETF market environment is constructed using News-Informed Financial Trend Yield (NIFTY) dataset (Saqr et al., 2024). It contains the processed and curated daily news headlines from 2019-07-29 to 2020-09-21 and generated sentiment categories for each news headline.

In experimental use, we divide the dataset according to the date, with the train set used for the warmup phase to establish the memory database, and the test set used for the test phase to evaluate the model performance.

3.3 Evaluation metrics

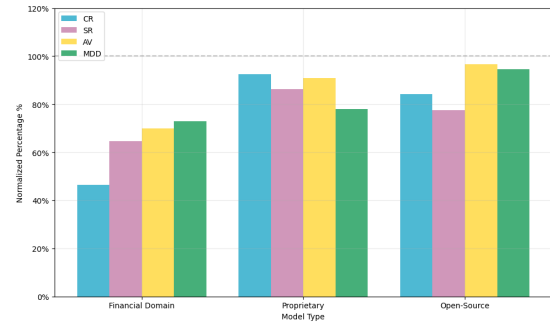
We employ four widely recognized financial metrics to evaluate and compare the investment performance of various LLMs serving as backbones across different tasks: : Cumulative Return (CR) (Hull, 2007), Sharpe Ratio (SR) (Sharpe, 1994),

Table 1: INVESTORBENCH evaluates 13 proprietary or open-source LLMs on financial decision-making tasks.

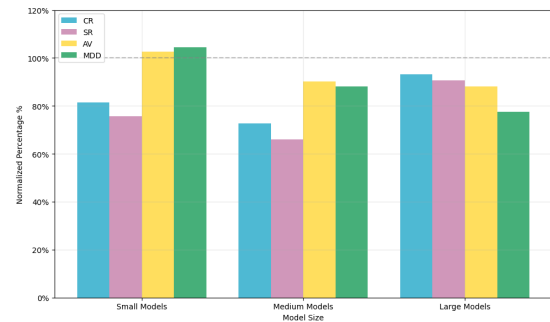
Model	#Size	Form	Ver.	Model	#Size	Form	Ver.
gpt-4t(Achiam et al., 2023)	N/A	api	0613	Qwen2.5-72b(Qwen team, 2024)	7B	open	Instruct
gpt-4o(OpenAI, 2022)	N/A	api	0806	Qwen2.5-32b(Qwen team, 2024)	32B	open	Instruct
gpt-4o-preview	N/A	api	0912	Qwen2.5-72b(Qwen team, 2024)	72B	open	Instruct
DeepSeek-v2(Xin et al., 2024)	15B	open	Lite	Llama3.1-8b(Llama team, 2024)	8B	open	Instruct
DeepSeek-VL2(Xin et al., 2024)	67B	open	Chat	Llama3.1-70b(Llama team, 2024)	70B	open	Instruct
Yi-1.5-9b(Young et al., 2024)	9B	open	Chat	Palmyra-Fin(team, 2024)	70B	open	32K
Yi-1.5-34b(Young et al., 2024)	34B	open	Chat				

Annualized Volatility (AV) (Cochrane, 1988), and Maximum Drawdown(MDD) (Ang and Chen, 2003). Note that CR and the SR are often considered more essential than AV and MDD in evaluating asset trading performance due to their focus on long-term gains and risk-adjusted returns by their definition. Here, we regard these two metrics as primary metrics when evaluating the experiment outcomes. The detailed explanation is in Appendix B.

4 Experiment and Discussion



(a) The performance comparison by different model types.



(b) By model parameter sizes across open-source LLMs. Note: Small-size models refer to models with no more than 10B parameters. Medium-size models refer to models with parameter numbers in the range of (10B, 65B]. Large-size models are those with more than 65B parameters.

Figure 3: Agent Performance Comparisons from two key perspectives. The CR, SR, AV, and MDD represent the average values for each model type, expressed as a percentage relative to the Buy & Hold strategy.

To establish a baseline and assess the performance of LLM agents, we standardize experimental settings and evaluation metrics across various financial decision-making tasks. Results are pre-

Table 2: Performance of stock trading with different LLMs as backbone model across seven stocks.

Model	MSFT				JNJ				UVV				HON			
	CR $\uparrow$	SR $\uparrow$	AV $\downarrow$	MDD $\downarrow$	CR $\uparrow$	SR $\uparrow$	AV $\downarrow$	MDD $\downarrow$	CR $\uparrow$	SR $\uparrow$	AV $\downarrow$	MDD $\downarrow$	CR $\uparrow$	SR $\uparrow$	AV $\downarrow$	MDD $\downarrow$
Buy & Hold	15.340	1.039	24.980	9.428	13.895	1.343	17.500	9.847	36.583	2.112	29.299	15.406	33.256	2.347	23.967	9.195
<i>Financial Domain Models</i>																
Palmyra-Fin-70B	14.697	0.897	27.518	9.428	5.748	0.450	19.317	9.367	37.875	2.039	31.200	15.967	20.016	1.464	22.974	6.824
<i>Proprietary Models</i>																
GPT-o1-preview	17.184	0.962	30.000	9.428	13.561	1.086	20.864	9.847	41.508	2.147	32.479	9.633	13.162	0.776	28.511	11.558
GPT-4	16.654	0.932	30.022	9.428	13.712	1.103	20.894	9.860	31.791	1.640	32.567	10.434	34.342	2.005	28.779	9.195
GPT-4o	12.461	0.924	22.653	6.647	9.099	0.875	17.471	7.169	8.043	0.496	27.241	14.889	38.540	2.418	26.782	8.979
<i>Open-Source Models</i>																
Qwen2.5-72B-Instruct	7.421	0.588	21.238	6.973	14.353	1.140	20.995	9.812	37.178	1.822	34.223	13.365	34.309	2.000	28.779	9.292
Llama-3.1-70B-Instruct	17.396	1.335	21.892	7.045	13.868	1.121	20.779	9.825	35.981	1.728	34.986	15.406	43.944	2.646	27.903	8.993
DeepSeek-67B-Chat	13.941	0.834	28.081	7.850	14.426	1.185	20.450	9.825	29.940	1.481	33.964	15.407	32.536	1.909	28.628	10.782
Yi-1.5-34B-Chat	22.093	1.253	29.613	9.428	14.004	1.180	19.938	9.847	20.889	1.020	34.417	14.936	30.743	1.823	28.335	9.195
Qwen2.5-32B-Instruct	-0.557	-0.041	22.893	8.946	2.905	0.292	16.725	7.169	-1.623	-0.097	27.973	17.986	26.332	1.980	22.348	5.261
DeepSeek-V2-Lite (15.7B)	11.904	0.694	28.796	16.094	-7.482	-0.670	18.773	17.806	33.560	1.703	33.099	12.984	16.686	0.974	28.771	16.806
Yi-1.5-9B-Chat	19.333	1.094	29.690	9.428	18.606	1.611	19.409	10.986	49.415	2.410	34.446	11.430	29.028	1.700	28.682	12.588
Llama-3.1-8B-Instruct	22.703	1.322	28.855	7.385	13.988	1.486	20.460	9.969	41.108	1.981	34.866	16.429	39.079	2.320	28.299	10.341
Qwen-2.5-Instruct-7B	-10.305	-0.724	23.937	23.371	21.852	0.980	37.425	9.573	11.752	0.853	22.988	15.451	4.291	0.285	24.933	14.156
<i>Financial Domain Models</i>																
Palmyra-Fin-70B	-6.661	-0.222	50.379	25.820	8.562	0.372	38.891	25.466	-3.261	-0.057	101.711	58.406	-0.453	0.031	63.660	36.564
<i>Proprietary Models</i>																
GPT-o1-preview	34.499	0.796	72.822	35.490	8.238	0.422	33.045	14.412	32.433	0.558	102.470	54.016	25.057	0.592	69.446	34.639
GPT-4	45.246	1.190	63.896	25.031	9.889	0.440	38.008	19.119	75.952	1.286	104.059	37.867	43.696	0.972	68.654	27.339
GPT-4o	45.946	1.348	57.281	21.631	7.405	0.457	27.434	12.824	63.743	1.318	85.210	29.220	39.031	1.041	56.642	21.225
<i>Open-Source Models</i>																
Qwen2.5-72B-Instruct	39.112	1.075	61.136	26.985	11.935	0.572	35.293	19.119	87.412	2.181	70.628	12.464	46.153	1.276	55.686	19.523
Llama-3.1-70B-Instruct	37.545	0.891	70.815	29.813	12.772	0.583	37.057	16.021	66.522	1.118	104.848	46.379	38.946	0.864	70.907	30.738
DeepSeek-67B-Chat	35.647	0.885	67.660	33.359	14.213	0.666	36.118	10.876	30.963	0.599	91.146	45.855	26.941	0.717	64.975	30.030
Yi-1.5-34B-Chat	35.364	0.808	73.561	35.490	14.227	0.623	38.596	19.432	64.307	1.063	106.597	48.042	37.966	0.831	72.918	34.321
Qwen2.5-32B-Instruct	21.336	0.729	49.157	20.704	13.220	1.089	20.522	8.943	28.096	0.652	72.344	37.975	20.884	0.823	47.341	22.541
DeepSeek-V2-Lite (15.7B)	31.458	0.744	68.524	35.404	27.016	1.221	11.860	37.435	27.762	0.474	103.193	48.478	28.745	0.813	61.192	40.439
Yi-1.5-9B-Chat	31.350	0.703	74.895	37.975	3.640	0.162	37.947	17.578	33.748	0.569	104.502	55.284	22.913	0.478	72.448	36.946
Llama-3.1-8B-Instruct	35.622	0.832	71.936	36.383	7.079	0.309	38.757	18.747	33.689	0.560	106.094	56.527	25.463	0.567	72.262	37.219
Qwen-2.5-Instruct-7B	41.203	0.925	74.862	37.975	14.336	0.694	34.924	12.029	33.007	0.546	106.437	53.054	29.515	0.722	72.074	34.353

¹ The Buy & Hold strategy is a passive investment approach commonly used as a baseline strategy, where an investor purchases stocks and holds onto them for an extended period regardless of market fluctuations.

² An upward arrow ($\uparrow$) next to a metric indicates that higher values signify better performance, while a downward arrow ($\downarrow$) indicates that lower values are preferable.

³ The numbers highlighted in red indicate the best-performing outcomes for the corresponding metrics.

sented on a task-by-task basis. We report the performance of INVESTORBENCH on three single-asset trading tasks: *stocks*, *cryptocurrencies*, and *ETFs* trading, using closed-source, open-source, and domain-specific LLMs.

4.1 Experiment Setup

Table 1 summarizes the performance of a comprehensive list of trading agents. For single equity tasks, the baseline is set up by Buy and Hold strategy, while for portfolio management task, it is set up by an equal-weight portfolio with the detailed rationale explained in Appendix. In our experiments, the temperature parameter of all LLM-based agent systems is set at 0.6 to balance response consistency and reasoning creativity. The performance metrics are reported for the test trajectory with the median CR, SR, AV, and MDD from five repeated epochs. (If the median of these metrics does not belong to the same epoch, the performance is based on the trajectory with the median SR.)

Furthermore, the selection of warm-up and test periods differs across various tasks due to the varying time spans of data collected to construct the

agent environment. For the single-asset trading tasks, the warm-up period of stock trading is from 2020-07-01 to 2020-09-30 and the test period is from 2020-10-01 to 2021-05-06. The warm-up period of cryptocurrency trading is from 2023-02-11 to 2023-04-04 and the test period is from 2023-04-05 to 2023-11-05. The warm-up period of ETF trading is from 2019-07-29 to 2019-12-30 and the test period is from 2020-01-02 to 2020-09-21.

For LLM deployment, we utilize vllm to deploy LLMs. For small-scale LLMs (under 10B parameters), we deploy models on two RTX A6000 GPUs, each with 48GB DRAM. For mid-scale LLMs (10B to 65B parameters), we use four RTX A6000 GPUs. For large-scale LLMs (over 65B parameters), models are deployed on eight A100 GPUs, each equipped with 80GB DRAM.

4.2 Result 1: Stock Trading

Table 2 presents the performance of thirteen backbone models across seven stocks, accompanied by the average of each metric for all stocks to offer a more comprehensive view of their overall performance. We outline three key insights as follows: